

Alsurvey.me v2.0 — Feature Implementation Guide

Why-Protocol Engine

Overview

The Why-Protocol is a conversational AI technique that automatically generates follow-up “why” questions to extract tacit knowledge. When a respondent gives a surface-level answer, the engine probes deeper.

Implementation

Location: `lib/ai-helper.ts` → `generateWhyFollowUp()`

How it works:

1. User answers a survey question
2. AI analyzes the response for depth and knowledge signals
3. If the response contains actionable knowledge but lacks explanation, the Why-Protocol activates
4. A contextual follow-up is generated: “You mentioned X — why did you choose that approach over alternatives?”
5. This continues up to `whyProtocolDepth` levels (configurable per survey, default: 3)

Configuration:

```
// In Survey model
whyProtocolDepth: 3 // 1-5, higher = deeper probing
domainTemplate: 'sap-knowledge' // Adds domain-specific context
```

Domain Context:

Each domain template injects specialized context into the AI prompt:

- SAP: module names, transaction codes, customization terminology
- Healthcare: clinical protocols, patient safety, evidence-based practice
- IT Ops: ITIL framework, incident management, SLA terminology
- Government: policy frameworks, regulatory compliance, stakeholder mapping

Customization

To add a new domain template:

1. Create `lib/question-flows/your-domain.ts`
2. Define question flow with domain-specific questions
3. Add domain context strings for AI prompts
4. Register in `lib/question-flow.ts` → `getFlowForTemplate()`

Tacit Knowledge Score (TK-Score)

Overview

A multi-dimensional scoring system that quantifies the tacit knowledge value of each survey response.

Dimensions

Dimension	Weight	What it measures
Depth	40%	How deep the knowledge goes (surface vs. expert-level)
Uniqueness	35%	How rare/unique the knowledge is
Actionability	25%	How directly applicable the knowledge is

Calculation

Location: `lib/ai-helper.ts` → `calculateTacitKnowledgeScore()`

```
composite = (depth * 0.4) + (uniqueness * 0.35) + (actionability * 0.25)
```

The AI evaluates each dimension on a 0-100 scale based on:

- **Depth signals:** Technical detail, cause-effect reasoning, edge cases mentioned
- **Uniqueness signals:** Non-obvious insights, personal experience, counter-intuitive findings
- **Actionability signals:** Step-by-step knowledge, decision criteria, concrete examples

Integration Points

- Stored on `Response` model after conversation completion
- Displayed in admin dashboard TK-Score tab
- Used to identify expert profiles (high TK-score → potential expert)
- Published as Knowledge-OS events for cross-module use

Knowledge Graph

Overview

Automatic extraction of knowledge entities and relationships from survey responses, stored as a graph structure.

Architecture

Location: `lib/knowledge-graph.ts`

```
Survey Response → AI Entity Extraction → KnowledgeNode + KnowledgeEdge → Graph Storage
```

Node Types

- **CONCEPT:** Abstract ideas, methodologies, frameworks
- **SKILL:** Specific competencies and abilities
- **PROCESS:** Workflows, procedures, sequences

- **TOOL:** Software, hardware, instruments
- **PERSON:** Individuals (anonymized) with specific knowledge
- **DECISION:** Key decision points and their rationale
- **EXPERIENCE:** Specific experiences that shaped knowledge

Edge Types

- **RELATES_TO:** General association
- **DEPENDS_ON:** Prerequisite relationship
- **LEADS_TO:** Causal or sequential relationship
- **CONTRADICTS:** Conflicting knowledge
- **ENABLES:** One thing makes another possible
- **REQUIRES:** Mandatory dependency
- **PART_OF:** Hierarchical containment

Future Visualization

The admin dashboard includes a placeholder for graph visualization. Recommended libraries:

- **D3.js** — Full control, complex but powerful
- **Cytoscape.js** — Purpose-built for graphs
- **React Flow** — React-native graph rendering

Expert Profile System

Overview

Automatic identification of knowledge experts based on survey responses and TK-scores.

Risk Assessment

Risk Level	Criteria	Action
LOW	Expert has documented their knowledge	Monitor
MEDIUM	Partial documentation, knowledge transferable	Plan transfer
HIGH	Critical knowledge, limited documentation	Urgent: schedule knowledge capture
CRITICAL	Single point of failure, no backup	Emergency: immediate capture + cross-training

Identification Logic

1. Response TK-Score > 75 → Flag respondent as potential expert
2. Multiple high-score sessions → Confirm expert status
3. Domain extraction from responses → Map expertise areas
4. Risk assessment based on uniqueness scores and domain coverage

Knowledge-OS Integration

Event-Driven Architecture

Location: `lib/knowledge-os-integration.ts`

AIsurvey.me → Event Bus → EDI / learning.me

Event Types

Event	Trigger	Consumers
KNOWLEDGE_CAPTURED	Survey response completed	EDI, learning.me
EXPERT_IDENTIFIED	Expert profile created/up-dated	EDI, HR systems
GRAPH_UPDATED	New nodes/edges added	Knowledge Graph UI
INSIGHT_GENERATED	AI analysis completed	learning.me, reports

Integration Configuration

Set in `.env` :

```
KNOWLEDGE_OS_WEBHOOK_URL=https://your-knowledge-os.com/webhook
KNOWLEDGE_OS_API_KEY=your-api-key
```

Or configure via API:

```
curl -X POST /api/admin/integration \
  -d '{"action": "configure", "module": "edi", "config": {"webhookUrl": "...",
"apiKey": "..."}'}
```